
INSTRUCTIONS TO THE STUDENTS:-

- 1) Question 1 is compulsory.
 - 2) Attempt any five questions from Q 2 to Q 7.
 - 3) Draw neat labeled diagram wherever necessary.
 - 4) Assume suitable data wherever necessary.
-

Q.1 Objective type questions.

- 1 Each action potential in the heart originates near the top of the right atrium at a point called _____. (01)
 - 2 EEG frequency alpha band range is : _____. (01)
 - 3 The term _____ refers to any component of the signal that is extraneous to the variable represented by the signal. (01)
 - 4 Heat, excitement or local inflammation can cause _____ of capillaries, which increases the blood flow, at least locally. (01)
 - 5 An indocyanine dye named _____, used in an isotonic solution was a long favoured indicator. (01)
 - 6 The interconnections between the neurons are called as _____. (01)
 - 7 The _____ is simply a device that carries sound energy from the chest of the patient to the ear of the physician via a column of air. (01)
 - 8 Blood platelets are having diameter in range of ____ to _____. (01)
 - 9 The _____ wave in ECG represents depolarization of atrial musculature. (01)
 - 10 For DC currents, threshold of perception of electric shock is about _____ to _____ mA. (01)
-

- Q.2 A) What is a delay line? Explain its importance. (04)
- B) State various factors encountered in measuring a living system. (04)
- C) Define
- a) Repolarization of a cell. (04)
 - b) Depolarization of a cell.
 - c) Absolute refractory period.
 - d) Relative refractory period.
- Q.3 A) Explain the indirect method of blood pressure measurement using sphygmomanometer. (06)
- B) Write a note on neuronal receptors. (06)
- Q.4 A) Explain the working of CAT scanners with neat diagram. (06)
- B) Explain the properties of ultrasound. (06)
- Q.5 A) What are central monitors? Discuss the advantages of using Central Monitors? (06)
- B) State the special situations of electric shock hazards in hospitals. (06)
- Q.6 A) State and explain the principle of Linear strain gauge. (06)
- B) State the factors deciding the choice of particular transducer in biomedical instrumentation. (06)
- Q.7 A) Explain the principle of Flame Photometer with neat diagrams. (06)
- B) Explain the conductivity coulter method of blood cell counting. (06)
-